

$$p = \left(\frac{E}{c}, \vec{p} \right)$$

$$\underline{F} = \left(\gamma \frac{d}{dt} \left(\frac{E}{c} \right), \gamma \frac{d\vec{p}}{dt} \right) = \left(\gamma \frac{\vec{F} \cdot \vec{v}}{c}, \gamma \vec{F} \right)$$

Newtonov zakon

$$\underline{F} = m \frac{d\underline{v}}{dt}$$

po komponentah:

$$\frac{1}{c} \frac{dE}{dt} = mc \frac{d\gamma}{dt} = mc \gamma^3 \frac{\vec{v} \cdot \vec{a}}{c^2}$$

$$E = mc^2$$

$$\frac{d\gamma}{dt} = \frac{d}{dt} \left(1 - \frac{v^2}{c^2} \right)^{-\frac{1}{2}} = \left(-\frac{1}{2} \right) \gamma^3 \left(\frac{2\vec{v}}{c^2} \right) \frac{d\vec{v}}{dt} = \gamma^3 \frac{\vec{v} \cdot \vec{a}}{c^2}$$

$$\frac{d\vec{p}}{dt} = m \left(\frac{d\gamma}{dt} \vec{v} + \gamma \frac{d\vec{v}}{dt} \right) = m \left(\gamma^3 \frac{\vec{v} \cdot \vec{a}}{c^2} \vec{v} + \gamma \vec{a} \right)$$

$$\underline{F} = \gamma \left(mc \gamma^3 \frac{\vec{a} \cdot \vec{v}}{c^2}, m \gamma \vec{a} + m \gamma^3 \frac{\vec{v} \cdot \vec{a}}{c^2} \vec{v} \right) \quad / : m$$

$$\underline{a} = \frac{\underline{F}}{m} = \gamma \left(\gamma^3 \frac{\vec{a} \cdot \vec{v}}{c}, \gamma \vec{a} + \gamma^3 \frac{\vec{v} \cdot \vec{a}}{c^2} \vec{v} \right) \quad \text{"2. Newtonov zakon" v relativ. obl.}$$

$$\text{v lim } \underline{a} \rightarrow (0, \vec{a}) \quad \checkmark$$

Zgled: gibanje delca v konst. $\vec{E}$
(delec z nabojem e posp. iz
mirovanja s poljem $\vec{E}$)

$$m d(\gamma v) = F dt = e E dt \quad / : S$$

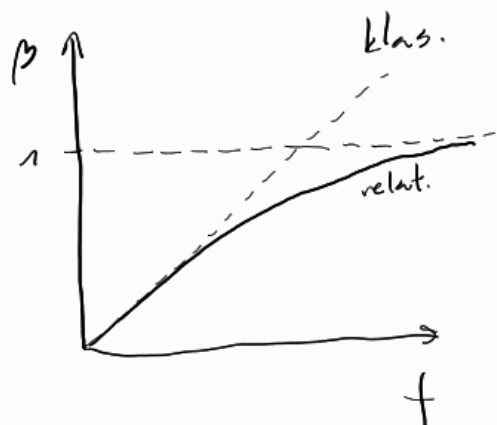
$$\text{dobimo, } \rightarrow m \gamma v = e E t$$

$$\text{ker } v_0 = 0 \quad \beta \gamma = \frac{e E t}{mc} = \alpha t \rightarrow \frac{\beta}{\sqrt{1-\beta^2}} = \alpha t \rightarrow \beta = \frac{v}{c} = \frac{\alpha t}{\sqrt{1+\alpha^2 t^2}}$$

$$\gamma = \sqrt{1+\alpha^2 t^2}$$

$$T = \int_0^t \frac{dt}{\gamma} = \int_0^t \frac{dt}{\sqrt{1+\alpha^2 t^2}} = \frac{1}{\alpha} \sinh^{-1}(\alpha t) \rightarrow \boxed{t = \frac{1}{\alpha} \sinh(\alpha T)}$$

koord. čas se povečuje eksponentno hitreje kot T



$$x(t) = c \int_0^t \beta(t) dt = c \int_0^t \frac{\alpha t}{\sqrt{1+\alpha^2 t^2}} dt = \frac{mc^2}{eE} (\sqrt{1+\alpha^2 t^2} - 1) \rightarrow \lim_{t \ll \frac{1}{\alpha}} = \frac{1}{2} \frac{eE}{m} t^2 = \frac{1}{2} a t^2$$

(majhni zasi)

- krajevno spremenljiva polje $\vec{E}$:

$$A = \int \vec{F} \cdot d\vec{s} = e \int \vec{E} \cdot d\vec{s} = eU_{\text{potenc. razlika}}$$

če iz mirovanja:

$$\Delta E_k = eU$$

$$mc^2(\gamma - 1) = eU$$

$$\gamma = \frac{eU + mc^2}{mc^2}$$

$$\beta^2 = 1 - \frac{m^2 c^4}{(eU + mc^2)^2} \rightarrow \beta = \sqrt{\frac{eU(eU + 2mc^2)}{(eU + mc^2)^2}} = \begin{cases} \sqrt{\frac{2eU}{mc^2}}; & eU \ll mc^2 \\ 1; & eU \gg mc^2 \end{cases}$$

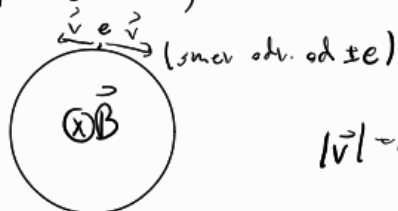
$$v = \sqrt{\frac{2E_k}{m}}; E_k = eU = \frac{mv^2}{2}$$

za $eU \ll mc^2$

↑ se giblje skoraj s c

- delec v konst. mag. polju

$$\vec{F} = e\vec{v} \times \vec{B}, \perp \text{ na smer gibanja (delec kroži)}$$



$|\vec{v}| = \text{konst.}$

$$\rightarrow m \frac{d(\gamma \vec{v})}{dt} = m \gamma \frac{d\vec{v}}{dt} = e\vec{v} \times \vec{B}$$

- ker imamo gibanje le v ravnini: $a_r = \omega_c^2 r = v \omega_c = \frac{v^2}{r}$

$$1) m \gamma \omega_c v = e v B$$

$$\boxed{\omega_c = \frac{eB}{m\gamma}} \text{ ciklotronska frekvenca}$$

$$2) m \gamma \frac{v^2}{r} = e v B$$

$$r = \frac{m \gamma v}{eB} = \frac{p}{eB} \rightarrow \boxed{p = eBr} \text{ velja tudi relat.}$$

Zgled:

$$e^-, E = 5 \text{ GeV}, \gamma = \frac{E}{m c^2} \approx 10^4 \rightarrow p \approx \frac{E}{c} \text{ (ker } m c^2 \ll E^2)$$

$$B = 1 \text{ T}$$

$$\rightarrow r = \frac{p}{eB} \approx 17 \text{ m}$$

$$\omega_c \approx 1,8 \cdot 10^7 / \text{s}$$